MIL-C-46271A (MU)
8 February 1968
SUPERSEDING
MIL-C-46271 (Ord)
29 August 1960

MILITARY SPECIFICATION
CARTRIDGE, AIRCRAFT EJECTION SEAT, CATAPULT, M28A1 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Aircraft Ejection Seat, Catapult, M28A1, Assembly for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification, to the extent specified herein:

SPECIFICATIONS
Military

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-E-5272	-	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	-	Coatings, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS
Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-413	-	Visual Inspection Guide for Rubber O-Rings
MIL-STD-639	-	Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	-	Ammunition Data Card
MS-28741	-	Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

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DRAWINGS

D8593353	- Cartridge, Aircraft Ejection Seat, Catapult, M28A1, Assembly
F8797308	- Carton, Packing, Ammunition for Cartridge, Aircraft Ejection Seat, Catapult, M28A1
F8797303	- Box, Packing, Ammunition for Cartridge, Aircraft Ejection Seat, Catapult, M28A1
IEL 8593353	- Inspection Equipment List for Cartridge, Aircraft Ejection Seat, Catapult, M28A1

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of any component.

3.3 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.4 Physical properties.- Physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.- The cartridge when assembled in an M5A1 Catapult shall propell a 300 pound weight vertically and shall comply with the requirements of Table I.

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3.5.2 (Cont'd)

TABLE I

Temperature Degree(°) Fahrenheit(F)	Minimum Velocity ft/sec (See 6.1)	Acceleration, Max g's Pressure Time	Acceleration, Max g's Load Cell	Max Rate of Change of Acceleration g/sec	Ignition Delay Milliseconds (See 6.2)
-65° to -70°F	45	16	14	100	150 max
65° to 75°F	60	20	18	170	150 max
160° to 165°F	64	24	24	250	150 max

3.5.2.1 Initiation.- The catapult shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 10 feet of aircraft hose, size 4, MS-28741 at 70° ± 5°F.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

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4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 23 cartridge cases, 23 cartridge heads, 50 units from each heat treated lot, 30 assembled cartridges for ballistics and 5 for sealant application to be tested in accordance with Table II and 10 sets of unassembled components (inclusive of explosive components) for visual and dimensional examination.

TABLE II

Paragraphs Requirement	Paragraphs Test	Test	Sample Size	Type of Test Dest	Type of Test Non-Dest
3.2	4.4.1	Vibration	12 (from Ballistics)	--	X
Dwg.	4.4.2	Airtightness	each	--	X
3.3	4.4.3	Protective finish	23 Ctg. Cases	X	--
			23 Ctg. Heads	X	--
3.4	4.4.4	Hardness	50	X	--
Dwg.	4.4.5	Sealant	5	X	--
3.5.2	4.4.6.1	Ballistics	30	X	--

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and conform to requirements as indicated in Table I. The sample for firing shall consist of the following:

TABLE III

Temperature	Non-Vibrated	Vibrated
-65°F to -70°F	6	4
65°F to 75°F	6	4
160°F to 165°F	6	4

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, a lot shall consist of 1200 cartridges, maximum, exclusive of test samples, and shall consist of:

a. Components produced by a homogeneous process.
b. Propellant from not more than one lot.
c. Primers from not more than one lot.
d. Igniter from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class, the criteria in Table V shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1-300	100% Examination	-	-	-	-
301-1300	110	1	2	2	3
1301-3200	150	2	3	3	4
3201-8000	225	3	4	4	5
8001-22000	300	4	5	5	6

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TABLE V

Total Allowable Defects Per Sample Size

Number of defects in class	Sample Size for Major Defects:			
	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over - 43	6	8	11	14

Number of defects in class	Sample Size for Minor Defects:			
	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over - 48	8	11	17	20

4.3.2.1 Primer, No. 33, Dwg. C7553919, covering a detail of Dwg.D859335

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Primer foil discolored	Visual
3. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

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4.3.2.2 Cartridge Case, Dwg. C8593354, covering a detail of Dwg.D8593353.

CRITICAL

None defined

Defect	Method of Inspection
MAJOR	
101. Concentricity of mouth with body	Gage
102. Straightness of base, min.	Gage
103. Diameter of body, min.	Gage
104. Diameter of mouth, min.	Gage
105. Thickness of base, min.	Gage
106. Thickness of sidewall, min.	Gage
107. Metal defective	Visual
108. Protective finish inadequate or defective	Visual
109. Burr	Visual
110. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of mouth, max.	Gage
203. Total length	Gage
MINOR	
204. Thickness of sidewall, max.	Gage
205. Location of datum diameter	Gage
206. Radius or chamfer missing or incorrect	Visual
207. Surface finish improper	Visual

4.3.2.3 Igniting Disc, Dwg. B8593355, covering a detail of Dwg. D8593353.

CRITICAL

None defined

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4.3.2.3 (Cont'd)

MAJOR

	Defect	Method of Inspection
101.	Diameter, min.	Gage
102.	Metal defective	Visual
103.	Protective finish inadequate or defective	Visual
104.	Burr	Visual
105.	Foreign matter	Visual

MINOR

201.	Diameter, max.	Gage
202.	Thickness	Gage

4.3.2.4 Firing Plug, Dwg. B8593498, covering a detail of Dwg. D8593353.

CRITICAL

1.	Length of striker, tip, min.	Gage

MAJOR

101.	Diameter of body, min.	Gage
102.	Profile of striker tip	Gage
103.	Height of boss	Visual
104.	Metal defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

MINOR

201.	Diameter of body, max.	Gage
202.	Diameter of boss	Gage
203.	Total length	Gage
204.	Length of striker tip, max.	Gage
205.	Surface finish improper	Visual

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4.3.2.5 Sealing Disc, Dwg. C8593494, covering a detail of Dwg. D8593353.

CRITICAL

	Defect	Method of Inspection
	None defined	

MAJOR

101.	Diameter, min.	Gage
102.	Marking missing, incorrect or illegible	Visual
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

MINOR

201.	Diameter, max.	Gage
202.	Thickness	Gage
203.	Varnish missing	Visual

4.3.2.6 Cartridge head, Dwg. C8593608, covering a detail of Dwg.D8593353.

CRITICAL

None defined

MAJOR

101.	Concentricity of primer pocket recess with primer pocket	Gage
102.	Concentricity of recess on top of head with primer pocket	Gage
103.	Concentricity of flange on head with body	Gage
104.	Diameter of primer pocket	Gage
105.	Diameter of primer pocket recess, min.	Gage
106.	Diameter of body, min.	Gage
107.	Diameter of flange, min.	Gage
108.	Depth to bottom of primer pocket recess	Gage
109.	Depth of primer pocket to chamfer at base	Gage
110.	Location of datum diameter	Gage
111.	Metal defective	Visual
112.	Protective finish inadequate or defective	Visual
113.	Surface finish of primer pocket, primer pocket recess and top of head, improper	Visual
114.	Burr	Visual
115.	Foreign matter	Visual

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4.3.2.6 (cont'd)

	Defect	Method of Inspection
MINOR		
201.	Diameter of primer pocket recess, max.	Gage
202.	Diameter of recess on top of head	Gage
203.	Diameter of center hole	Gage
204.	Diameter across center boss	Gage
205.	Diameter across cut-out	Gage
206.	Diameter across spring retaining groove	Gage
207.	Diameter of body, max.	Gage
208.	Diameter of step at spring retainer groove	Gage
209.	Diameter of flange, max.	Gage
210.	Depth of recess on top of head	Gage
211.	Depth of cut-out	Gage
212.	Depth of lip of cut-out	Gage
213.	Depth to lip cut-out	Gage
214.	Total length	Gage
215.	Width of undercut	Gage
216.	Location of center holes	Gage
217.	Radius or chamfer missing or incorrect	Visual
218.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.7 Retaining Ring, Drawing B8593499 covering a detail of Drawing D8593353.

CRITICAL

None defined

MAJOR

101.	Concentricity of center hole with body	Gage
102.	Diameter of body, min.	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of center hole	Gage
202. Diameter of body, max.	Gage
203. Thickness	Gage

4.3.2.8 Retaining Spring, Drawing B8593610, covering a detail of Drawing D8593353.

CRITICAL	
None defined	
MAJOR	
101. Thickness of wire, min.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter free	Gage
202. Thickness of wire, max.	Gage
203. Gap, min. (when spring is compressed to specified O.D.)	Gage

4.3.2.9 Cartridge, Aircraft Ejection Seat, Catapult, M28A1, Assembly Drawing D8593353.

CRITICAL	
1. Any part missing	X-ray & Visual 1/
2. Improper assembly	X-ray & Visual 1/
3. Primer not properly seated	Visual
4. Marking missing, incorrect or illegible	Visual

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4.3.2.9 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Concentricity of head with body	Gage
102.	Depth of recess on head	Gage
103.	Length of head	Gage
104.	Crimp missing or improper	Visual
105.	Sealant missing	Visual
106.	Sealant in cavity	Visual
107.	Foreign matter	Visual
MINOR		
201.	Diameter, max.	Gage
202.	Diameter of head, max.	Gage
203.	Total length	Gage
204.	Location datum diameter	Gage
205.	Varnish missing or inadequate	Visual

1/ The cartridge shall be X-rayed with the sides of the case perpendicular to the rays. The X-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.10 Carton, Packing, Ammunition for Cartridge Aircraft Ejection Seat, Catapult, M28A1, Drawing F8797308.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.10 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Contents loose	Manual & Visual
202.	Quantity of contents incorrect	Visual
203.	Separator or filler missing	Visual
204.	Heat sealer incomplete, loose or distorted	Visual
205.	Cut or damage through impregnated box	Visual
206.	Marking missing, incomplete or illegible	Visual

4.3.2.11 Box, Packing, Ammunition for Cartridge, Aircraft Ejection Seat, Catapult, M28A1 Drawing F8798303.

CRITICAL

None defined

MAJOR

None defined

	MINOR	
201.	Contents loose	Manual & Visual
202.	Hardware missing, loose, broken or improperly engaged	Visual
203.	Strapping missing, loose, broken or improperly assembled	Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of cartonsincorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection Testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Vibration.- Twelve of the thirty cartridges of the initial production sample shall be subjected to vibration. The twelve cartridges shall then be X-rayed. Failure of any cartridge to comply with the requirements of 3.2 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of twelve cartridges for vibration tests.

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4.3.3.2 Airtightness. Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.3 Protective Finish.-Twenty-three units of each component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.3.3.4 Hardness.-Fifty units from each heat treated lot of components shall be tested. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the lot of components.

4.3.3.5 Sealant Application.-Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.6 Functional.-The following test shall be performed by the Government.

4.3.3.6.1 Ballistics.-Thirty assembled cartridges from the initial production sample and thirty cartridges from each subsequent production lot shall be tested. Failure of any cartridge to comply with the requirements of 3.5.2 shall be cause for rejection of the lot.

4.3.4 Inspection Equipment.-The examination and tests shall be made using equipment listed on IEL-8593353.

4.4 Test Methods and Procedures

4.4.1 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.2 Airtightness.-The method for determining airtightness shall be by use of equipment listed on IEL-8593353.

4.4.3 Protective Finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and cartridge head shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.4 Hardness.-A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

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4.4.5 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Functional

4.4.6.1 Ballistics.- The cartridges shall be loaded into an acceptable M5A1 Catapult, divided into three equal groups and temperature conditioned in accordance with Table III. The cartridge shall be tested within ten minutes after removal from the conditioning chamber. When this time is exceeded the catapult shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.- A standard M5A1 catapult loaded with a dummy M28A1 Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATIONS FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawings F8797308 and F8797303.

5.2 Data card.-Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Velocity applies at tube separation.

6.2 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

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Preparing Activity

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Project No. 1377-A092

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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